

Experiment No. 6

Title:

Demonstration of String Methods and Regular Expressions (RegEx); Development of a Text Processing Tool using JavaScript.

Software / Tools Required:

1. Visual Studio Code (VS Code)
2. Google Chrome
3. HTML5
4. CSS3
5. JavaScript (ES6)

Theory:

String methods and regular expressions are used in JavaScript to process, search, and validate text.

(a) String Methods

JavaScript provides various methods to perform operations on strings, such as:

1. `split()` – splits a string into an array.
2. `indexOf()` – finds the position of a specified text.
3. `replace()` – replaces matching text.
4. `reverse()` – reverses an array.
5. `join()` – joins array elements into a string.

(b) Regular Expressions (RegEx)

Regular expressions are patterns used to search, match, and validate text. They are commonly used for checking email addresses, extracting information, and performing text substitutions.

(c) Text Processing

Text processing involves analyzing a paragraph, counting words and vowels, searching for specific text, replacing words, and reversing the entered paragraph.

(d) Email Validation

Regular expressions are used to check whether an entered email address follows a valid format.

Text Processing Tool

Accepts User Input Such As:

1. Paragraph
2. Email Address

It Performs:

1. Counts the number of words.
2. Counts vowels using regular expressions.
3. Replaces specified text.
4. Finds the position of a word.
5. Validates an email address.
6. Extracts email addresses from text.
7. Reverses the entered paragraph.

Result / Conclusion:

The practical was completed successfully. The concepts of **string methods and regular expressions** were implemented. A text processing tool was developed using JavaScript that analyzes text, performs string operations, validates email addresses, and displays the results accurately.